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POCUS-Emergency

Intelligent Multimodal Assistant for Clinical Ultrasound
and Decision Support in Emergencies

A multimodal decision-support system combining patient information, three ultrasound modules and a clinical reasoning layer — with an explicit account of what it does not know.

4

perception models

339

automated tests

37

safety properties

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The problem

Emergency assessment is multimodal, and the modalities arrive apart

1

Time pressure

Decisions are made before all the information is available.

2

Scattered evidence

Vitals, complaint, laboratory values and imaging arrive separately.

3

Operator dependence

POCUS adds real information at the bedside — its interpretation varies with the operator.

4

Integration

Combining these under time pressure, while tracking what is still unknown, is where error enters.

The gap

**No system combines
imaging with the rest
of the emergency record
while keeping an honest
account of its own
reliability.**

That gap is the subject of this work.

The objective is assistance, not replacement. The physician remains responsible for the clinical decision.

The proposed solution

One workflow, several sources of evidence, one decision

Centralised evidence

Triage, three POCUS modules and the clinical record on one screen.

Explicit uncertainty

Every finding carries its reliability, and absent evidence is labelled absent — never normal.

Structured reasoning

The assessment is assembled from enumerated evidence, not narrated freely.

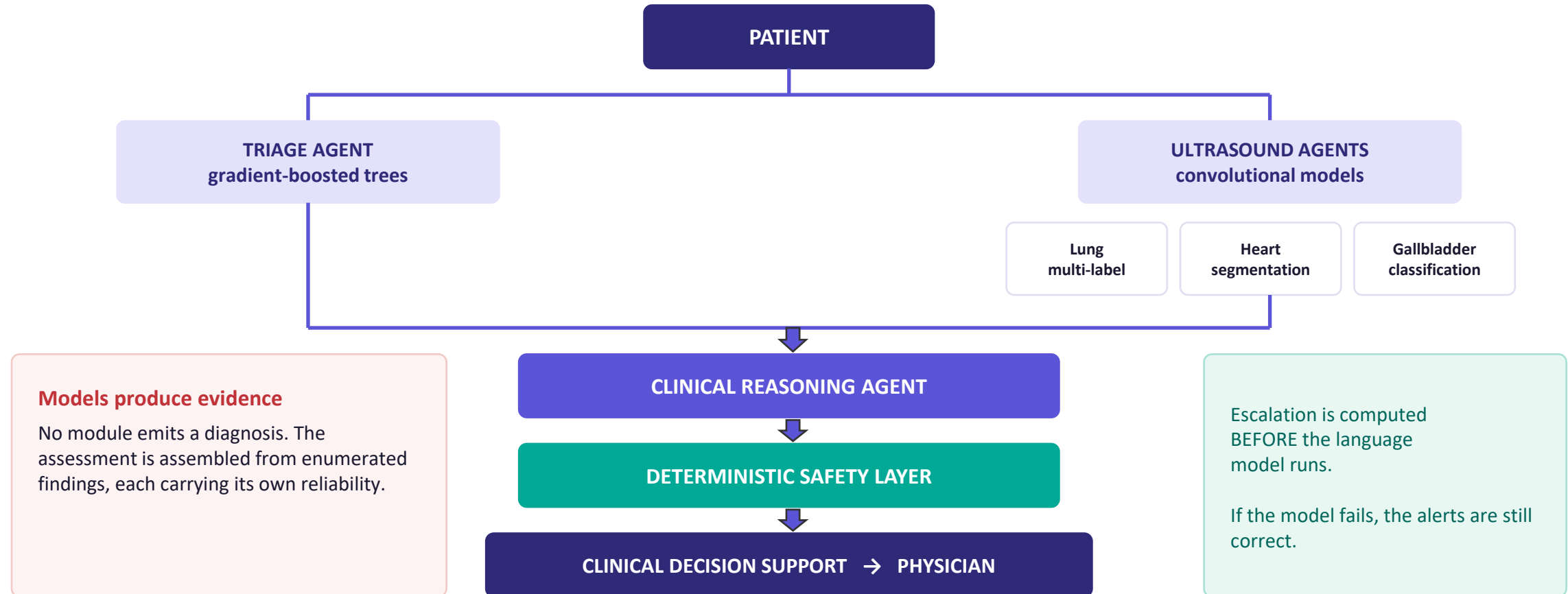
Persistent record

Each physician keeps their own patients, examinations and images.

Delivered as a working web platform with accounts, a stored patient history, containerised deployment and an automated test suite.

Technical architecture

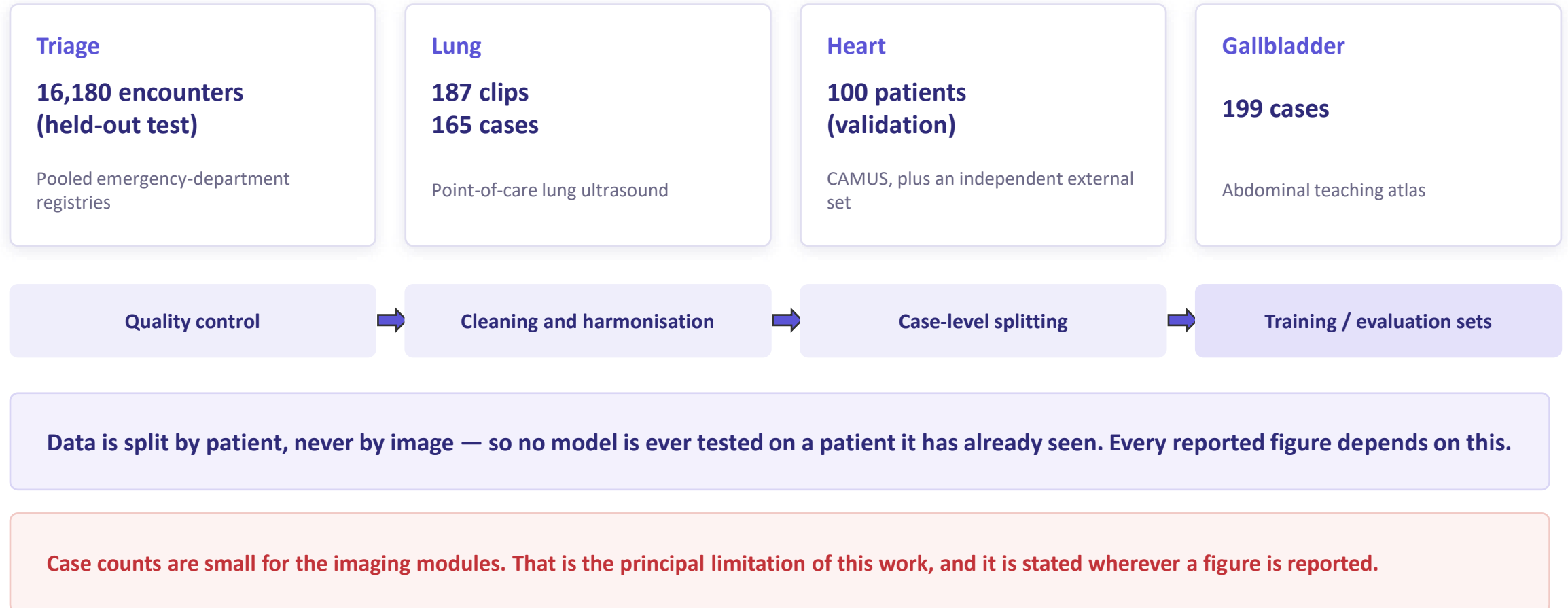
CRISP-DM · Modeling



Data sources and preparation

CRISP-DM · Phases 2–3

Four independent public sources, prepared under one rule



Module 1 — Triage

Modeling

Clinical task

Grade the urgency of a patient from information available at the bedside.

Data

Pooled emergency-department registries. Disposition fields removed after they were found to leak the outcome into the input.

Approach

Gradient-boosted trees on 21 prepared inputs — 10 entered by the clinician, 11 derived (shock index, pulse pressure, red-flag indicators). Probabilities calibrated.

Output

A suggested urgency tier with its probability — presented to the clinician, who confirms or changes it. The suggestion is never applied on its own.

The deployed model was retrained on exactly the fields the interface collects, so its reported accuracy is accuracy a bedside prediction can actually reach.

Results — Triage

CRISP-DM · Evaluation

Held-out test set, 16,180 encounters

63.0%

Accuracy

56.7%

Macro F1

77.3%

High-urgency recall

1.36%

Dangerous miss rate

Two models, and why the weaker one is deployed

The reference model also sees hospital registration fields and reaches 68.0% accuracy. The deployed model uses only what a clinician types at the bedside: 63.0% accuracy, and a LOWER dangerous-miss rate — 1.36% against 1.55%. On the error that matters, the deployable model is the better one.

A dangerous miss is a genuinely high-acuity patient graded low. These are retrospective figures on registry data: a decision-support suggestion, not a validated triage instrument.

Model selection: gradient-boosted trees were compared against logistic regression, random forest, a multilayer perceptron and CatBoost on identical features and splits. None beat them — the ceiling is a property of the data.

Module 2 — Lung

Modeling

Clinical task

Detect four findings on point-of-care lung ultrasound: B-lines, consolidation, pleural effusion, pleural thickening.

Data

187 clips from 165 cases. Frames sampled per clip; clips of one case never split across folds.

Approach

EfficientNet-B0 backbone with a multi-label head — findings co-occur, so they must not compete for probability mass. Three-fold grouped cross-validation, per-finding thresholds.

Output

Per-finding detection with a confidence, reduced across frames by median so one blurred frame cannot carry the result.

Pneumothorax is deliberately NOT modelled: it is a motion sign, invisible in a single static frame, and the dataset contains no positive cases.

Results — Lung

Pooled out-of-fold, 187 clips from 165 cases

Finding	AUROC	Sensitivity	Positive cases
B-lines	0.830	84.1%	63
Consolidation	0.832	60.0%	60
Pleural effusion	0.718	75.0%	24
Pleural thickening	0.773	65.2%	46

Macro AUROC 0.788

Pleural effusion rests on 24 positive cases, and per-fold AUROC for B-lines ranges from 0.536 to 0.830. A pooled figure conceals that spread, which is why the spread is reported.

Pneumothorax is not modelled and therefore cannot be excluded by this system. It is reported as not assessed — never as absent.

Calibration was not applied to this module: a held-out calibration split would leave about three positive effusion cases, and a curve fitted on three points is not a calibration. The interface states that the confidence is uncalibrated.

Module 3 — Heart

Modeling

Clinical task

Segment the left ventricle and derive an ejection-fraction band from the segmentation.

Data

CAMUS, 100 validation patients, plus an independent external collection used for validation only and never for training.

Approach

U-Net with an EfficientNet-B0 encoder, four output classes. Ejection fraction computed from end-diastolic and end-systolic areas, with a geometric volume correction.

Output

Left-ventricle segmentation and an ejection-fraction band — severe, reduced or normal — with the value flagged when it falls close to a cutoff.

Segmentation quality and clinical output are reported separately. The band is what a clinician acts on, so the band is the module's result.

Results — Heart

CRISP-DM · Evaluation

Segmentation is strong; the derived clinical band is weaker, and that is the figure reported

0.928

LV Dice, in-domain
(nnU-Net: 0.93)

0.534

LV Dice, external

63.4%

EF-band balanced acc.
(chance 33.3%)

6.9 pp

EF mean absolute error
(was 12.2 pp)

Ejection fraction here is a geometric estimate derived from segmentation — a proxy, not a definitive echocardiographic measurement. Volume correction halved its error.

Values near a band boundary are flagged

A 6.9-point error is enough to move a patient across a cutoff. Where an estimate falls within that margin of a threshold, the system states the value could lie on either side. The margin is the module's own measured error, not a round number chosen for convenience.

The external drop was first attributed to a labelling-convention mismatch. Connected-component analysis over the full external set disproved that hypothesis, and the reversal is reported rather than dropped.

Module 4 — Gallbladder

Modeling

Clinical task

Identify the principal abnormality in a gallbladder study, among five reported groups.

Data

199 cases from an abdominal teaching atlas. One class was excluded after per-case inspection showed its contents did not match its label.

Approach

EfficientNet-B0 trained on eight fine-grained classes, reported as five by summing group probabilities. Temperature scaling for calibration.

Output

One reported finding with a calibrated confidence. Groups are mutually exclusive: classes considered and rejected are not passed on as evidence.

The atlas contains no healthy class. This module can say which abnormality is most consistent with the image — it cannot say a gallbladder is normal.

Results — Gallbladder

CRISP-DM · Evaluation

Five reported groups, held-out evaluation

59.7%

Balanced accuracy
(chance 20.0%)

0.024

Calibration error
after temperature scaling

199

Cases

Why the taxonomy matters more than the accuracy

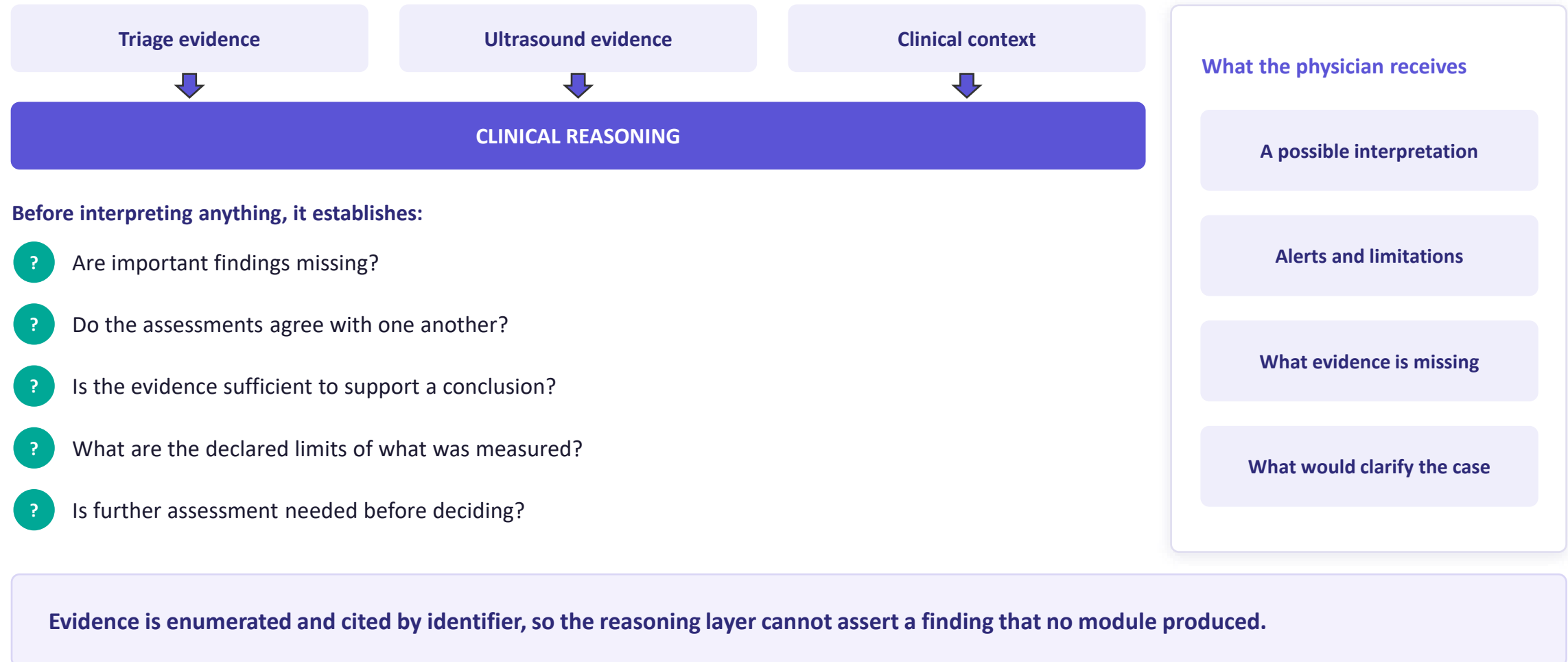
82% of the model's confusions cross a reported group boundary. A coarser three-category grouping was tested and rejected on that evidence: if the groups captured a distinction the model could see, its confusions would concentrate within them. They do not.

No healthy class. The model reports which abnormality is most consistent with the image; it cannot report the absence of disease, and the interface says so beside every result.

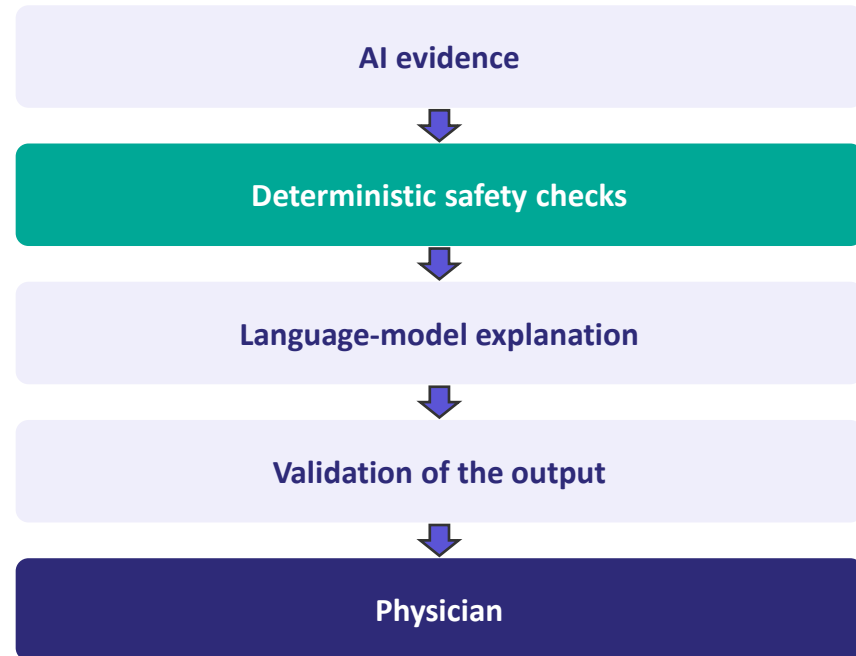
A ninth folder present in the original release was excluded after sampling one frame per case across the whole set. A first inspection of a handful of frames had judged it consistent — frames from one case are not a sample of a class.

The Clinical Reasoning Agent

How the branches are combined



Safety architecture



Computed before any
model runs:

Missing evidence
Disagreement
Severity
Uncertainty

Four states a finding can occupy

 **detected**
a positive finding

 **unreliable**
detected, weak evidence

 **screened, not detected**
looked for, not found

 **not assessed**
no model covers this

Absence of evidence is never recorded as evidence of absence. This is enforced by the data structure, not by a convention someone must remember.

Global validation

CRISP-DM · Evaluation

What was verified, and how

339

Automated tests
all passing

37

Safety properties
each reported separately

150

Regression checks
over 10 full cases

6 / 6

Automated checks
tests, lint, security

Grouped by property, not by count

- Conflict detection — 14 / 14
- Absent is not normal — 24 / 24
- Hallucination rejection — 25 / 25
- Access control — 17 / 17

"339 passed" is not evidence. "Conflict detection 14/14" is.

Beyond model metrics

- Generalisation tested on an independent set where one existed
- Calibration measured per module, and refused where data was insufficient
- Whole encounters run end to end, intake through report
- Individual real studies inspected by hand

Seven well-motivated changes were tested and rejected on measurement rather than preference — each reported with its outcome.

Technical deployment

CRISP-DM · Deployment

A running application, not a notebook

1

Application

FastAPI and uvicorn serving a generated clinician interface in English and French

2

Containerised

Docker image running as a non-root user, with a health check that reports whether each model actually loaded

3

Accounts and storage

Per-physician accounts, hashed passwords, server-side sessions, and a stored patient record

4

Automated pipeline

Every check runs on each push: tests, regression, lint, code and dependency security

Technical deployment and clinical deployment are different claims. The first is demonstrated here; the second — installation in a hospital and prospective validation — has not been attempted.

The clinician interface

The interface is titled "POCUS-Emergency Clinical Copilot". It features a sidebar on the left with navigation options: Home, Patient workup, Diagnosis, Patient record, Assessment (selected), Alerts (7), Clinical assistant, Timeline, Report, and History. The main content area is titled "Clinical assessment" for "Patient A · 78 · F · shortness of breath". A "Generate clinical report" button is in the top right. A red alert box at the top states: "HIGH CLINICAL PRIORITY. Acute dyspnoea presentation assessed at HIGH severity. Point-of-care ultrasound detected b lines (lung, 0.82), pleural effusion (lung, 0.58). NOT ASSESSED: heart — expected for this presentation and never examined. 3 critical alert(s): spo2 88.0 % is below the configured critical bound of 90.0 %; a positive imaging finding with troponin, lactate never measured; expected but never assessed: heart — a gap in the record, not a negative result. ESCALATED — 2 trigger(s): route simulation. Most informative next: heart ultrasound, troponin, lactate." A "Review alerts" button is next to it. Below the alert, the interface is divided into three columns. The left column, "What the assistant sees", lists symptoms (shortness of breath), vital signs (hr 118 bpm, sbp 96 mmHg, rr 26 /min, spo2 88 %, temp 37.4 C, dbp 58 mmHg, pain 4 /10), POCUS findings (b lines — Detected (0.82), pleural effusion — Detected — weak evidence (0.58), consolidation — Screened, not detected (0.05), pneumothorax — NOT assessed — not modelled (—)), and laboratory results (all "Not measured"). The middle column, "Differential assessment", states "No differential was generated. Producing one requires a 4.9 GB language model on a GPU, which is not loaded in this deployment. Everything else on screen was computed here." The right column, "What would help clarify this case?", lists tests not performed: heart ultrasound, troponin, lactate, bnp, d_dimer, and crp. A "CURRENT PATIENT" box at the bottom left shows "Patient A" with "78 · F · shortness of breath" and a "HIGH" alert. A bottom bar shows "AR Perception modules" for lung, heart, and gallbladder.

POCUS-Emergency Clinical Copilot

Search patients, findings, reports...

Assistant ready Français + New assessment

Clinical assessment

Patient A · 78 · F · shortness of breath

HIGH CLINICAL PRIORITY
Acute dyspnoea presentation assessed at HIGH severity. Point-of-care ultrasound detected b lines (lung, 0.82), pleural effusion (lung, 0.58). NOT ASSESSED: heart — expected for this presentation and never examined. 3 critical alert(s): spo2 88.0 % is below the configured critical bound of 90.0 %; a positive imaging finding with troponin, lactate never measured; expected but never assessed: heart — a gap in the record, not a negative result. ESCALATED — 2 trigger(s): route simulation. Most informative next: heart ultrasound, troponin, lactate.

What the assistant sees
The information available for this patient.

SYMPTOMS
shortness of breath

VITAL SIGNS
hr 118 bpm
sbp 96 mmHg
rr 26 /min
spo2 88 %
temp 37.4 C
dbp 58 mmHg
pain 4 /10

POCUS
b lines — Detected (0.82)
pleural effusion — Detected — weak evidence (0.58)
consolidation — Screened, not detected (0.05)
pneumothorax — NOT assessed — not modelled (—)

LABORATORY
troponin — Not measured
bnp — Not measured
d_dimer — Not measured
lactate — Not measured
crp — Not measured
wbc — Not measured
creatinine — Not measured
ph — Not measured

Differential assessment
Offered for consideration. This is not a diagnosis.

No differential was generated. Producing one requires a 4.9 GB language model on a GPU, which is not loaded in this deployment. Everything else on screen was computed here.

What would help clarify this case?
These tests were not performed. Their absence does not mean they are normal.

- heart ultrasound**
expected for this presentation and never assessed; an unperformed scan is missing information, not a negative result
- troponin**
named by an escalation trigger as a key value that was never measured
- lactate**
named by an escalation trigger as a key value that was never measured
- bnp**
never measured; absent rather than normal
- d_dimer**
never measured; absent rather than normal
- crp**
never measured; absent rather than normal

Generate clinical report

Review alerts

CURRENT PATIENT
Patient A
78 · F · shortness of breath
• HIGH

AR Perception modules
• lung • heart • gallbladder

On one screen

- Patient information
- Triage suggestion
- Ultrasound findings
- What was not measured
- Alerts and limitations
- What would clarify the case

Every value on the screen is computed by the pipeline. Where the system has nothing real to put in a slot, the slot says so.

Served in English and in French — the French text is composed from the same computed fields, so a French screen cannot state something the English record does not.

What this could bring to an emergency department



Centralised evidence

Triage, imaging and laboratory values assembled in one place rather than read separately.



Faster access to findings

A POCUS study is read and filed against the patient as it is acquired.



Explicit uncertainty

Every finding carries its reliability, and what was not measured is named.



Structured reasoning support

A written assessment with its alerts, its gaps and what would clarify the case.



Persistent history

Each physician's patients, examinations and images kept across visits.



Extensible

Further organs and findings slot into the same contract and the same safety layer.

The value proposition is the workflow, not any single model. No individual module reaches the accuracy a standalone diagnostic tool would require.

Limitations and roadmap

Stated plainly — a system that hides these is not usable clinically

Current limitations

- Small datasets for several ultrasound findings
- External generalisation remains difficult (LV Dice 0.928 → 0.534)
- Ejection fraction is a geometric proxy
- Pneumothorax is not assessed at all
- The gallbladder model has no healthy class
- No prospective clinical validation

Roadmap

- Larger and more diverse clinical datasets
- External and prospective validation
- Improved robustness of the ultrasound modules
- Evaluation with emergency physicians
- Additional organs within the same safety contract
- Integration into a real clinical workflow

The system is built to accept validated data later: versioned by content hash, retrained, evaluated against fixed criteria, and approved by a named person before any model reaches a patient.

What it can tell you — and what it cannot

It can

assemble the available evidence, grade urgency, read three organ systems, and state what is missing.

It cannot

exclude a pneumothorax, confirm a normal gallbladder, or replace an echocardiographic measurement.

It never

makes the clinical decision. That remains yours.

339 automated tests · 37 safety properties · 150 regression checks · 4 perception models · a working platform

Thank you · Questions